

Polar Coordinates

Key Points

Polar coordinates describe a point using a **distance** and an **angle**

$$(r, \theta)$$

Remember $r \geq 0$, and either $-\pi \leq \theta \leq \pi$ or $0 \leq \theta \leq 2\pi$

To convert from Cartesian (x, y) to polar coordinates (r, θ) use

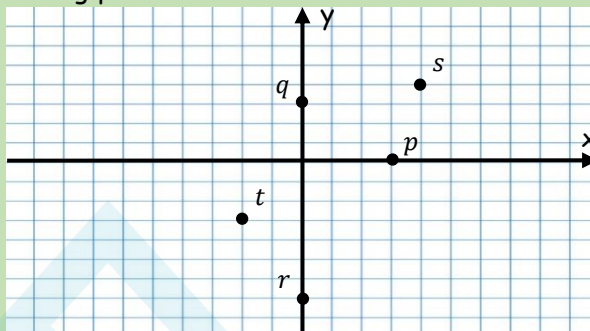
$$x = r \cos(\theta) \text{ and } y = r \sin(\theta)$$

To convert from polar to Cartesian coordinates use

$$r^2 = x^2 + y^2 \text{ and } \tan(\theta) = \frac{y}{x}$$

Basic Skills

Q1. Using the diagram below, write the polar coordinates of the following points



- a) p b) q c) r d) s e) t

Competency Skills

Q2. Convert the following Cartesian coordinates to polar coordinates

- a) (3, 4) b) (-5, 12) c) (-3, -3) d) (5, -1)

Q3. Convert the following polar coordinates to Cartesian coordinates

- a) $(\sqrt{2}, \frac{\pi}{4})$ b) $(2, \frac{3\pi}{4})$ c) $(3, \frac{7\pi}{6})$ d) $(7, \frac{10\pi}{6})$

Q4. Convert the expression $\sqrt{x^2 + y^2}$ to polar coordinates

Q5. Convert the equation $r = \frac{5}{\sin(\theta)}$ into Cartesian coordinates in the form $y = f(x)$

Advanced Skills

Q6. Explain using a diagram why $x = r \cos(\theta)$

Q7. Explain the restrictions that are placed on r and θ

Q8. Convert the equation $y^2 = 4(1 - x)$ into polar coordinates

Q9. By converting the equation $r = \frac{3}{\sin(\theta) - 2 \cos(\theta)}$ into Cartesian coordinates, find the point of intersection with the line $x + y = 6$

Q10. Let A and B be the points given by the polar coordinates $(2, \frac{\pi}{2})$ and $(4\sqrt{2}, \frac{5\pi}{4})$ respectively. Find the points of intersection of the line through A and B , with the circle with $r = 3$

Extension

Q11. Find all points of intersection between the line $y = x$ and the spiral $\theta = r$

Hint: Split the line $y = x$ into two half-lines